

Functions and Recursion in C++

> **Definition to Remember**

> Fun
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1. Concept Overview & Fundamentals

Functions and Recursion in C++ provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

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2. Technical Architecture & Memory Mechanics

C++ Runtime Execution Architecture

```text

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### Modern C++ Feature Comparison

| Modern C++ Paradigm | Legacy Equivalent | Engineering Advantage | Performance Impact |

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3. Core Syntax & Implementation Patterns

Modern C++ Template Structure

```
```cpp
// Fast I/O and standard namespace configuration

std::cout << "Executing C++ Module: Function and Recursion" << "\n";

return 0;
```
```

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4. Real-World Applications & System Engineering

- **Quantitative Finance & HFT:** High-frequency trading platforms use modern C++ for microsecond-level execution.

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5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: add function

```
Problem Description & Objective:

C++ Code Implementation:
```cpp
// Implementation
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using namespace std;
```

```
int addFourNumbers(int a, int b, int c, int d) {
```

```
// Return sum of four numbers
```

```
// --- Main Execution Entry Point ---
```

```
int main() {
```

```
int sum = addFourNumbers(w, x, y, z);
```

```
// Output the sum
```

```
// Return zero code indicating successful program finish
```

```
return 0;
```

```
---
```

Problem 2: fibonacci recursion

****Problem Description & Objective:****

****C++ Code Implementation:****

/*

#include <iostream>

// Use standard library namespace globally

int fibonacci(int n) {

// --- Main Execution Entry Point ---

// Output formatted data to console stream

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 3: get average function

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
double get_average(int a, int b, int c, int d, int e) {
```

```
// --- Main Execution Entry Point ---

double avg = get_average(v1, v2, v3, v4, v5);

// Return zero code indicating successful program finish

```

#### Problem 4: greet function

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
void greet() {
```

```
// --- Main Execution Entry Point ---
```

```

```

### Problem 5: increment function

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
void incrementByValue(int n) {
```

```
void incrementByReference(int &n) {
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
incrementByValue(val);
```

```
incrementByReference(val);
```

```
// Return zero code indicating successful program finish
```

```

```

### Problem 6: max function

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>

// Use standard library namespace globally

float getMax(float a, float b) {
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 7: palindrome recursion**

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using namespace std;
```

```
int reverseRecursive(int num, int rev = 0) {
```

```
// Evaluate recursive logic
```

```
bool isPalindrome(int num) {
```

```
// Evaluate conditional logic
```

```
// --- Main Execution Entry Point ---
```

```
int main() {
```

```
// Evaluate conditional decision logic
```

```
if (isPalindrome(num))
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### Problem 8: print date function

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
using n
```

```
void print_date() {
```

```
time_t
```

```
// --- Main Execution Entry Point ---
```

```
int main
```

```
---
```

Problem 9: square function

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
int square(int n) {
```

```
// Retu
```

```
// --- Main Execution Entry Point ---
```

```
int main
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

## 6. Exam Scoring Points & Memory Revision

```
> Must-Write Points
```

```
> 1. C
```

```
> Quick Recall
```

```
> `Sig
```